CROSS: Recoverability-Aware Self-Distillation via Cross-Rollout Supervision Selection

Anonymous ACL submission

Abstract

Learning from generated reasoning requires deciding which corrective feedback should become supervision and where it should act. We study supervised-token efficiency: in-domain reasoning performance relative to the cumulative token uses that directly enter the training objective. We introduce CROSS, a cross-rollout supervision selection framework for language model self-distillation. Successful peers propose corrections to failed trajectories; cross-proposal corroboration selects supported repairs, and guided continuations assess whether the current student can use them. Accepted corrections condition a frozen initial self-teacher. Squared recovery rates weight forward-KL supervision over the precisely located suffix of the original student trajectory, while the training student receives ordinary task context. Experiments on science reasoning show the highest cross-field average among the compared methods under a shared protocol that selects each seed's best checkpoint after training. Complementary learning curves track mean performance against cumulative supervised tokens; their selected checkpoints combine higher scores with fewer supervised tokens than standard on-policy self-distillation across the evaluated fields. In Biology, CROSS learns rapidly with limited direct supervision, demonstrating the value of selective feedback for token-efficient learning. CROSS connects corrective evidence, behavioral usability, and supervision location to make better use of direct token supervision.

1 Introduction

Language models generate natural-language reasoning trajectories that expose both successful solutions and recurring failure states. These trajectories can supply training data through rationale bootstrapping (Zelikman et al., 2022) and on-policy distillation (Agarwal et al., 2024). On-policy self-distillation uses privileged training information to produce a self-teacher distribution along the student's own generated histories (Zhao et al., 2026a). This raises a practical question: how can corrective feedback be converted into effective token-level supervision without supervising every generated position?

We focus on *supervised-token efficiency*, relating in-domain reasoning performance to cumulative valid token uses in the training objective; we refer to this as token efficiency below. Repeated supervision counts repeatedly, and loss weights do not discount token counts. A useful correction must have supporting evidence, be usable by the current student, and identify where direct supervision should begin. A successful peer alone does not establish that every reasoning step is correct, and a supported repair may still be difficult for the student to execute. Supervision selection therefore needs to connect corrective evidence to the student's current reasoning state.

We introduce **CROSS** (**Cross-Rollout Supervision Selection**). For a failed trajectory, successful peers propose local corrections. CROSS compares proposals at the same error anchor, selects supported repairs, and maps their anchors to stored token boundaries. It then asks the current student to continue from the exact retained prefix under each correction. Task success measures conditional recoverability and determines signal admission and weight. A frozen initial self-teacher receives the accepted correction and authorized dataset information, while the training student retains ordinary task context. Forward KL acts on the original trajectory's valid suffix, weighted once by squared recoverability. Recovery continuations select supervision; the original rollout supplies the states on which it is applied.

Across SciKnowEval science fields, CROSS achieves the highest mean of per-seed best checkpoint scores in Chemistry, Biology, and Materials. It also leads the four-field average at 70.14%, com-

pared with 67.16% for OPSD. We report two complementary views: best in-domain performance, selected independently for each seed after training, and three-seed mean performance as a function of cumulative supervised tokens. The Biology curves show rapid learning under limited direct supervision. These trajectories reveal token efficiency separately from the best-checkpoint summaries. Across the four fields, the selected mean-curve checkpoints use substantially fewer supervised tokens than OPSD while reaching higher mean-curve scores. Section 4 specifies the reporting conventions.

Our contributions are:

- **Cross-rollout correction selection.** We construct failure-specific repairs and select existing proposals through cross-peer corroboration.
- **Recoverability-aware localized self-distillation.** Guided student continuations admit and weight corrections, whose teacher distributions supervise an exactly located suffix of the original trajectory.
- **An empirical study of token efficiency.** We combine full in-domain baseline comparisons with domain-specific performance–exposure curves, highlighting Biology's substantial learning gains under limited direct supervision.

2 Related Work

Self-distillation and privileged teachers. Rationale bootstrapping and rationale-based distillation use generated reasoning as supervision (Zelikman et al., 2022; Hsieh et al., 2023). Generalized distillation formalizes teacher access to information unavailable at deployment (Lopez-Paz et al., 2015), while GKD studies distillation on student-generated outputs (Agarwal et al., 2024). OPSD constructs a reference-conditioned self-teacher and matches its distributions along student rollouts (Zhao et al., 2026a); SDPO uses rich feedback or successful attempts (Hübotter et al., 2026). These methods establish the context asymmetry used by CROSS. Our selection object is a correction bound to a particular failed trajectory and error boundary.

Cross-rollout evidence and corrective supervision. ROSD contrasts a failed rollout with a successful peer to produce a corrective idea and an error quote for suffix-local distillation (Zhao et al., 2026b). MOPD conditions teachers on successful and failed peers (Yu et al., 2026b); GATES gates questions using consensus among privileged-context tutor answers (Stein et al., 2026). CROSS compares peer-conditioned proposals for the same error, selects a corroborated repair, and tests its usability through current-student recovery before distillation. This distinguishes its evidence and selection unit from peer-context construction and answer-level agreement.

Behavioral feedback and process supervision. SCoRe learns self-correction through reinforcement learning (Kumar et al., 2025), and CTRL evaluates critiques through the revisions they induce (Xie et al., 2025). Process supervision assigns labels to intermediate reasoning (Lightman et al., 2024); Math-Shepherd and VinePPO use sampled continuations for process annotation or credit assignment (Wang et al., 2024; Kazemnejad et al., 2024). CROSS measures continuation success conditioned on a specific correction and exact prefix. This conditional recovery statistic controls correction admission and weight in a distribution-matching objective.

Supervision selection and efficiency. Purified OPSD modifies the teacher distribution through contrasting contexts (Shen et al., 2026). PW-OPSD studies teacher-token reliability and applies position-dependent weights (Liu et al., 2026); EDGE-OPD uses context-induced probability changes for evidence masking (Lazaridis et al., 2026). TIP and SEAD select token positions using entropy and teacher–student disagreement or competence signals (Xu et al., 2026; Lee et al., 2026). CROSS instead selects the semantic correction and its supervision boundary, retaining dense distributional targets within an accepted suffix. We count directly supervised token uses separately from the generation and scoring needed to construct them. This resource distinction is essential when comparing correction selection with token selection or reward-driven updates. Extended discussion and a comparison by evidence source and selection unit appear in Appendix J.

CROSS uses corrections as the selection unit and the original rollout suffix as the supervised region. This connects corrective feedback to token efficiency: the empirical question is how much in-domain reasoning performance the selected super-

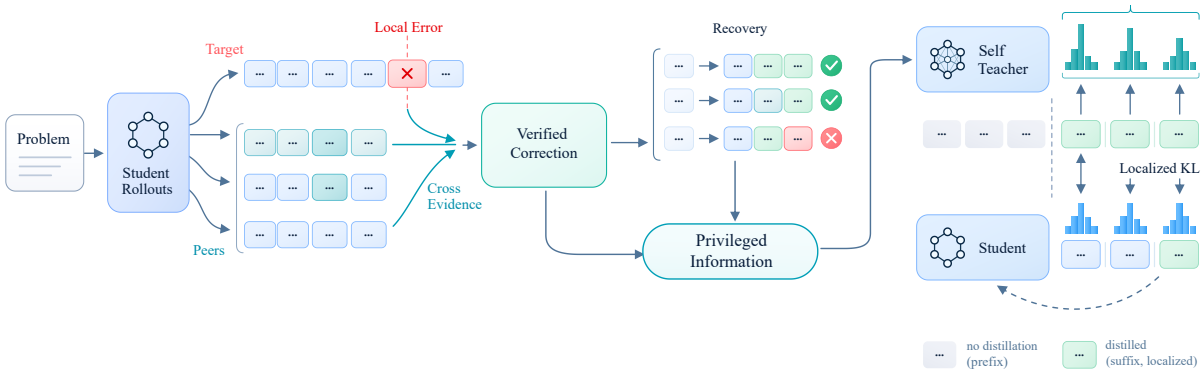


Figure 1: CROSS converts cross-rollout evidence into recoverability-aware, localized self-distillation. Successful peers guide corrections to a failed trajectory; the current student's guided continuations determine correction eligibility and weight. A frozen initial self-teacher then supplies soft targets along the original trajectory suffix. Prefix tokens provide context without direct loss. The recovery branches are schematic.

vision supports. Extended accounting and domain-specific curves distinguish the amount of direct supervision from the computation used to construct it.

3 Method

3.1 Problem Setup and Supervision Selection

CROSS constructs localized self-distillation supervision from the current student's unsuccessful reasoning trajectories. For each problem, it uses outcome-correct peer trajectories to propose corrections, selects corrections with cross-proposal corroboration, maps their proposed error onsets to exact stored token boundaries, and measures whether the same student can recover with this guidance. The resulting correction-level signals supervise the student against a frozen, information-augmented teacher.

At iteration t, let π_{θ_t} be the student and π_{θ_0} its fixed initial policy, used as the distillation teacher. For a public problem x, the student samples G rollouts. Each rollout retains its original completion token IDs, runtime availability, completion state, and task-verification outcome. The task checker distinguishes CORRECT, INCORRECT, and UNPARSEABLE outcomes; an unavailable assessment is recorded separately. Outcome labels identify successful peers and failed targets; correction selection uses the additional evidence defined below.

The selection object is a correction together with its target state and proposed error anchor. Cross-rollout corroboration determines whether a proposal is accepted, recovery measures the current student's response to that proposal, and distillation translates the accepted signal into a parameter update. We use *verified correction* for acceptance under the corroboration protocol and *exact localization* for the token mapping of its proposed anchor. Completion tokens are indexed from zero: for $\mathbf{y} = (y_0, \ldots, y_{T_y-1})$, $\mathbf{y}[:k]$ retains exactly k tokens. Figure 1 summarizes the information flow; Appendix B specifies notation and role-specific access.

3.2 Cross-Rollout Correction Construction and Verification

Target and peer evidence. Among available, complete rollouts with assessed outcomes, the target pool contains INCORRECT and UNPARSEABLE trajectories, and the peer pool contains CORRECT trajectories. Target and peer selection consumes typed outcomes without reading the checker-only reference answer.

To reduce surface redundancy, let $\phi(z)$ be the lexical token set of peer text z, obtained by Unicode normalization, case folding, and deterministic lexical tokenization. Peer overlap is measured by

$$s(z_i, z_j) = \frac{|\phi(z_i) \cap \phi(z_j)|}{|\phi(z_i) \cup \phi(z_j)|}. \quad (1)$$

Peers are connected when $s(z_i, z_j) \geq \tau_{\text{lex}}$, with $\tau_{\text{lex}} = 0.8$ in this procedure. Connected components define lexical-style clusters. We select one medoid from every cluster using the same similarity matrix, giving a deterministically ordered peer collection $\mathcal{Z} = (z_1, \ldots, z_K)$. Thus K is the number of clusters, without an additional upper cap. This construction reduces lexical redundancy among the evidence paths. Peer outcome correctness and lexical diversity supply complementary

selection criteria; neither requires that the sampled reasoning paths be independent.

A population with $K < 2$ produces no correction case. Otherwise, every eligible target is paired with the same peer collection. There is no target ranking or subsampling. This deterministic fan-out reuses the sampled peer evidence across all eligible failures. Normalization and medoid details appear in Appendix C.

Joint correction proposal. For a target $\mathbf{y}$, a deterministic structure-aware unitizer partitions its text into ordered reasoning units $(u_1, \ldots, u_L)$. Units preserve exact source substrings; protected mathematical and code blocks remain atomic. For each peer z_j, a declared generator $\mathcal{G}_\phi$ samples one proposed first-error unit a_j together with a correction c_j:

$$(a_j, c_j) \sim \mathcal{G}_\phi(\cdot \mid x, (u_1, \ldots, u_L), z_j), \quad j = 1, \ldots, K. \tag{2}$$

The model sees the public problem, numbered target units, and one complete peer trajectory. Dataset reference answers and checker diagnostics are not separate inputs; the complete peer may itself contain an answer. A structural parser checks the anchor and non-empty correction text. A failed pair removes only its own proposal. Successful proposals are grouped by identical anchor, retaining peer provenance, repeated text, and singleton groups. Proposal construction performs no semantic acceptance or majority filtering.

Pairwise corroboration. For a same-anchor group $\mathcal{C}_a = (c_1, \ldots, c_n)$ with $n \geq 2$, a relation model evaluates every unordered pair once in a fixed canonical order with greedy decoding. Its inputs are the public problem, target text through the proposed anchor, and the two corrections. Peer trajectories, reference answers, other pair judgments, and recovery outcomes are excluded. The relation r_{ij} is SUPPORT when the proposals share the same main actionable repair, OPPOSE when their main repairs are incompatible, and NOEVIDENCE otherwise. Different but compatible repairs can receive NOEVIDENCE. Missing assessments are kept distinct from this semantic label.

Signed graph selection. Each correction is a vertex; SUPPORT and OPPOSE create positive and negative edges, respectively. Neither NOEVIDENCE nor a missing assessment creates an edge. A *relation core* is a connected component of the graph formed by the union of positive and negative edges. Either edge type can connect vertices; a core need not contain both types. Within a core $\mathcal{H}$, define

$$\begin{aligned} d_j^+ &= \sum_{\ell \in \mathcal{H} \setminus \{j\}} \mathbb{I}[r_{j\ell} = \textsc{Support}], \\ d_j^- &= \sum_{\ell \in \mathcal{H} \setminus \{j\}} \mathbb{I}[r_{j\ell} = \textsc{Oppose}], \\ g_j &= d_j^+ - d_j^-. \end{aligned} \tag{3}$$

A core can emit a representative only from vertices with direct support:

$$j^\star_{\mathcal{H}} = \operatorname*{arg\,max}_{j \in \mathcal{H}:\, d_j^+ \geq 1} (g_j, d_j^+, -\operatorname{ord}(j)), \tag{4}$$

where the tuple is maximized lexicographically and $\operatorname{ord}(j)$ is stable upstream order. The selected text is an existing proposal and is never synthesized or rewritten. There is no additional positive-net-score, majority, or text-length requirement. A core without a supported vertex emits nothing; disconnected supported cores can yield multiple corrections for one anchor. Verification establishes cross-proposal support at the proposed anchor; behavioral usability is assessed by the subsequent recovery probe. Appendix D gives the reducer and missing assessment semantics.

3.3 Recoverability-Aware Supervision Selection

Conservative token boundary. For each selected correction, let $s^\star$ be the character onset of its proposed reasoning unit. We map this onset to the original stored completion IDs, using exact decoded character intervals. In raw decoded coordinates, let $\widetilde{s^\star}$ be the onset and $[b_r, e_r)$ the visible interval associated with token y_r. The retained prefix length is

$$k = \min\{r : b_r \leq \widetilde{s^\star} < e_r\}, \qquad P = \mathbf{y}[:k]. \tag{5}$$

If the onset lies inside a token, that whole token is excluded. If multiple token IDs contribute to the same Unicode character, the earliest contributor is excluded along with later tokens. This nearest-left projection preserves the original token sequence without crossing the proposed onset. The selected semantic anchor is preserved, with no maximum-retreat threshold. A correction cannot proceed when the required original IDs or exact mapping are unavailable.

Recovery under correction guidance. Recovery uses the exact student snapshot that generated $\mathbf{y}$, which is θ_t during the current logical batch. For correction c, it samples M new suffixes with the exact prefix supplied as assistant prefill:

$$\widetilde{U}^{(m)} \sim \pi_{\theta_t}(\cdot \mid x, c, P; \xi_m), \qquad m = 1, \ldots, M. \tag{6}$$

The recovery policy receives the correction but no separately supplied dataset answer. For the documented task interface, the objective checker evaluates the generated suffix directly. Correct, incorrect, and unparseable outcomes are assessed behavioral observations, including available continuations that reach their generation cap. Explicit infrastructure unavailability is excluded from the denominator. Let

$$\begin{aligned} A &= \sum_{m=1}^{M} \mathbb{I}[m \text{ is assessed}], \\ S &= \sum_{m=1}^{M} \mathbb{I}[m \text{ is correct}], \\ R &= \frac{S}{A} \quad (A > 0). \end{aligned} \tag{7}$$

A correction with $A = 0$ produces no recovery signal. Otherwise, eligibility requires $R \geq \tau_R$, and its supervision weight is $w = R^2$. The main experiments use $M = 7$ and $\tau_R = 4/7$. R measures task success conditional on the correction, retained state, and sampling protocol. It supplies a behavioral selection criterion rather than an estimate of the eventual gain from a gradient update. Appendix F specifies checkpoint binding, per-attempt randomness, and the recovery gate.

3.4 Localized Self-Distillation

Each recovery-qualified correction produces one *CrossSignal*, binding its problem, original target, selected correction, exact prefix boundary, and recovery score. The default online variant uses original-target scoring: for signal i, let $P_i = \mathbf{y}_i[: k_i]$, $U_i = \mathbf{y}_i[k_i :]$, and $Z_i = |U_i| > 0$. Successful recovery suffixes establish recoverability but do not replace U_i. Thus the supervised trajectory states come from the student's original rollout. The main method uses this original-target path; Appendix H records the separate trajectory variant.

Let C_i^S be the ordinary public problem context. The teacher context C_i^T additionally contains c_i and the dataset information authorized by the task profile. All four science fields use answer-only privilege. At each suffix position $u = 0, \ldots, Z_i - 1$, both policies are teacher-forced on the same exact observed token history:

$$\begin{aligned} q_{i,u}(v) &= \pi_{\theta_0}(v \mid C_i^T, P_i, U_i[: u]), \\ p_{i,u}(v; \theta) &= \pi_{\theta}(v \mid C_i^S, P_i, U_i[: u]). \end{aligned} \tag{8}$$

The student receives neither the correction nor the dataset answer as an explicit prompt input during distillation. This input boundary differs from the correction-conditioned recovery probe. Teacher parameters are frozen and teacher evaluation is gradient-free. The student retains gradients through prefix representations that influence supervised suffix predictions, although prefix positions receive no direct distillation loss.

Let $\mathcal{S}_t = (s_i)_{i=1}^{N_t}$ be the indexed collection of all eligible signals from the logical problem batch. For $N_t > 0$, CROSS minimizes

$$\mathcal{L}_t(\theta) = \frac{1}{N_t} \sum_{i=1}^{N_t} \frac{R_i^2}{Z_i} \sum_{u=0}^{Z_i - 1} D_{\mathrm{KL}}(q_{i,u} \,\|\, p_{i,u}(\cdot; \theta)) \,. \tag{9}$$

The divergence is teacher-to-student forward KL over the complete vocabulary, with distillation temperature one and uniform full-suffix weighting over valid completion positions. The inner mean normalizes within each signal; R_i^2 is then applied exactly once; the outer mean divides by the number of signals. We do not divide by the sum of recovery weights or rebalance by originating problem. Multiple corrections from one target or problem remain distinct signals when their provenance is distinct. Appendix G details causal alignment and physical gradient partitioning.

Online policy update. The student snapshot is held fixed while the full logical batch is sampled, processed, and recovery-assessed. All eligible CrossSignals and their normalization denominators are determined before accumulating the batch gradient. A non-empty signal batch produces one AdamW update; all student parameters are updated while the initial teacher remains fixed. When $N_t = 0$, data progress advances without changing student parameters or optimizer moments. Incomplete batches do not produce partial updates.

Each online signal contributes to at most one committed update. Revisiting a problem requires fresh evidence from the then-current student. After an update, the resulting student generates sub-

sequent raw rollouts and performs subsequent recovery probes. Here *on-policy* describes the origin of the stored rollout states and their snapshot-consistent use; the teacher has a different information context, and correction filtering changes the population that receives supervision. Algorithm 1 in Appendix A summarizes the complete training loop.

How selection determines the update. For student logit $\ell_{i,u,v}$ at a supervised position, the fixed teacher target gives

$$\frac{\partial \mathcal{L}_t}{\partial \ell_{i,u,v}} = \frac{R_i^2}{N_t Z_i}\big(p_{i,u}(v) - q_{i,u}(v)\big). \tag{10}$$

This identity makes the three selection decisions explicit. Corroboration and recovery determine membership in $\mathcal{S}_t$; localization determines the supervised positions; and R_i^2 scales the distributional correction within each signal. The $1/Z_i$ factor normalizes each suffix before signals are averaged, so suffix length does not by itself determine total signal weight. A shorter supervised suffix reduces the number of directly scored positions while preserving the retained prefix as context. These are properties of the objective; their effect on task performance is evaluated empirically. Appendix G derives the gradient and the equivalent reduction across physical microbatches.

The reduction also separates supervision exposure from optimization weight. Writing $\alpha_i = R_i^2/(N_t Z_i)$ for the coefficient of each retained position, the total coefficient assigned to signal i and to the batch is

$$\sum_{u=0}^{Z_i-1} \alpha_i = \frac{R_i^2}{N_t}, \qquad \sum_{i=1}^{N_t}\sum_{u=0}^{Z_i-1} \alpha_i = \frac{1}{N_t}\sum_{i=1}^{N_t} R_i^2. \tag{11}$$

Thus equally recoverable signals receive equal total loss coefficients even when their suffix lengths differ. At fixed signal count, a higher recovery rate increases a signal's coefficient quadratically. Because the outer normalization uses N_t, the mean recovery weight is retained in the batch objective; normalizing by $\sum_i R_i^2$ would remove this factor and define a different update. These coefficients characterize weighting, rather than the norm of the parameter gradient, which also depends on distributional disagreement and the model Jacobian. In contrast, supervised-token exposure counts the positions themselves, $\sum_i Z_i$, irrespective of these coefficients. This distinction motivates the accounting in Section 4.3.

4 Experiments

4.1 Setup

We evaluate in-domain performance and supervised-token efficiency through two complementary summaries: independently selected best checkpoints and three-seed learning curves. Each run trains on one SciKnowEval science field (Feng et al., 2024). Held-out evaluation contains 210 Chemistry, 50 Biology, 80 Physics, and 94 Materials Science questions. CROSS and all baselines share the same data splits. The metric mean@16 averages correctness over sampled responses and questions. Evaluation uses temperature 0.6, top-$p = 0.95$, top-$k = -1$, seed 42, and thinking disabled. Field averages receive equal weight.

The Qwen3-4B student (Yang et al., 2025) receives full-parameter updates at learning rate 10^{-5}; its initial policy remains frozen as teacher. Raw rollouts use temperature 1.3, top-$p = 0.95$, and top-$k = 20$. Recovery uses $M = 7$, $R = S/A$ over assessed attempts, and $R \geq 4/7$; accepted signals receive the original-target full-suffix objective in Eq. (9), with R^2 applied once. CROSS processes 500 batches of 16 questions and evaluates every 20 batches. All methods use seeds 42, 43, and 44. After training, we select the checkpoint with the highest test in-domain mean@16 independently for each seed, then report the mean and sample standard deviation of the three selected scores.

We compare CROSS with GRPO (Shao et al., 2024), SDPO (Hübotter et al., 2026), ROSD (Zhao et al., 2026b), MOPD (Yu et al., 2026b), and OPSD (Zhao et al., 2026a). Baselines use 200 optimizer steps. Table 1 reports the mean of per-seed best scores; the curves separately report three-seed mean performance versus cumulative supervised tokens. Detailed protocols and aggregation conventions appear in Appendix K.

4.2 In-Domain Performance

Table 1 compares the best performance attained by each run under the same post-hoc selection rule. CROSS achieves the highest mean in Chemistry, Biology, and Materials, while GRPO leads Physics. CROSS also has the highest four-field average at 70.14%, compared with 67.16% for OPSD and 69.71% for GRPO. Relative to OPSD, CROSS improves by 2.77, 2.92, 2.26, and 3.98 percentage points in the table's field order.

Method	Chemistry	Biology	Physics	Materials	Avg.
Base model	40.60	31.37	62.19	69.48	50.91
GRPO	75.23 ± 0.94	49.33 ± 10.68	$\mathbf{80.58} \pm 1.96$	73.68 ± 2.75	69.71
SDPO	75.68 ± 1.70	49.26 ± 0.98	67.76 ± 1.54	70.39 ± 1.20	65.77
ROSD	66.86 ± 17.95	51.50 ± 2.00	72.04 ± 2.15	74.14 ± 0.83	66.13
MOPD	50.05 ± 7.43	38.00 ± 7.49	70.21 ± 1.20	—	—
OPSD	73.75 ± 0.61	50.33 ± 1.34	71.12 ± 2.05	73.43 ± 0.99	67.16
CROSS	$\mathbf{76.52} \pm 2.16$	$\mathbf{53.25} \pm 3.27$	73.38 ± 2.13	$\mathbf{77.41} \pm 0.43$	**70.14**

Table 1: In-domain SciKnowEval mean@16 (%). For every method, the best test ID checkpoint is selected independently for each of seeds 42, 43, and 44 after training; entries report the mean $\pm$ sample SD of these three scores. Avg. equally weights all four fields and is omitted when a field is unavailable. Dashes indicate unavailable results. Bold marks the highest mean in each column.

Method	Chem.	Bio.	Phys.	Mat.
GRPO	31.01	18.31	27.16	10.17
SDPO	23.34	**3.80**	21.48	—
ROSD	**3.21**	4.55	16.81	3.49
MOPD	6.72	19.57	23.97	—
OPSD	32.72	27.19	34.06	24.61
CROSS	7.49	5.35	**7.92**	**2.59**

Table 2: Cumulative supervised tokens (M) at the best checkpoint on each method's three-seed mean curve. Bold marks the smallest available count per field. Dashes indicate that no post-initial checkpoint recovers the initial score. These curve-based selections are separate from the per-seed maxima in Table 1.

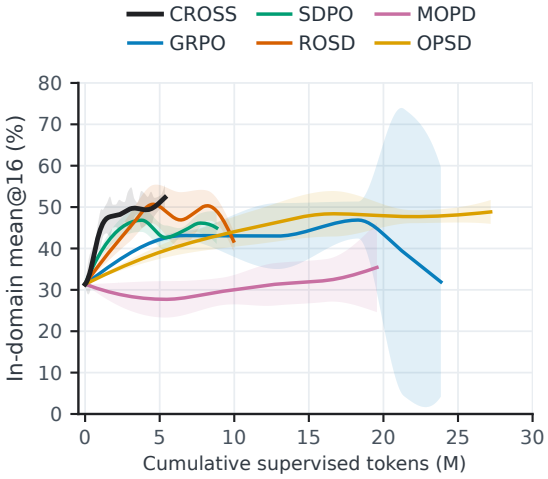


Figure 2: Biology in-domain mean@16 versus cumulative supervised tokens. Curves show the mean over seeds 42, 43, and 44 at corresponding token budgets. Shaded bands represent one standard deviation across seeds. Table 1 separately averages each seed's best checkpoint score.

Biology. Biology has the lowest base-model score in this comparison. CROSS reaches $53.25 \pm 3.27\%$, compared with OPSD's $50.33 \pm 1.34\%$ and ROSD's $51.50 \pm 2.00\%$. This summarizes the best scores attained independently by the three runs. The checkpoints need not occur at the same training step or cumulative token count.

4.3 Token Efficiency

A supervised token use is one valid position entering a committed training objective. Repeated uses count repeatedly, and R^2 does not discount counts. For the main objective, each update contributes $\sum_{i \in \mathcal{S}_t} Z_i$; prefix and masked positions contribute zero. Appendix K.2 details this accounting. These counts measure direct supervision; they exclude generation, verification, recovery, and teacher-scoring computation.

Supervision at selected curve checkpoints. Table 2 summarizes the token exposure at the best checkpoint of each method's three-seed mean trajectory. CROSS uses 7.49M, 5.35M, 7.92M, and 2.59M tokens for Chemistry, Biology, Physics, and Materials, respectively. OPSD uses $4.37\times$, $5.08\times$, $4.30\times$, and $9.49\times$ as many tokens at its selected curve checkpoints. At these selections, CROSS also has higher mean-curve scores than OPSD in all four fields. The comparison therefore links reduced direct supervision to strong learning outcomes across the science domains.

CROSS has the smallest available count in Physics and Materials. ROSD uses the fewest tokens in Chemistry, and SDPO uses the fewest in Biology. These counts describe where each method attains its own selected performance; the accompanying curves show the level of performance reached at that exposure.

Figure 2 describes the learning trajectory as supervision accumulates. CROSS improves rapidly

in the low-token region and reaches strong Biology performance with substantially less direct supervision than OPSD. At the selected curve checkpoints, CROSS reaches 52.29% at 5.35M tokens, while OPSD reaches 48.83% at 27.19M. Biology thus combines an improvement in the selected mean-curve score with a substantially smaller supervision count. The full trajectory complements this summary by showing the rapid early increase in CROSS performance.

The two summaries answer different questions. The table measures the best performance attained per seed within each training run; a curve measures the mean performance at a corresponding cumulative token budget. Averaging independently selected maxima need not produce a point on the mean curve. We therefore analyze the table's best scores and the curves' token efficiency separately, without assigning the table scores to curve coordinates.

Appendix K.3 presents the other three fields, extending this analysis across Chemistry, Physics, and Materials Science. Chemistry shows a sustained early increase for CROSS, while Physics includes longer improvements for GRPO. In Materials, CROSS reaches a strong mean score within a short supervision range. Together, the fields show why both the learning trajectories and their selected-checkpoint summaries are informative. Separate recovery and localization diagnostics appear in Appendix I.

5 Conclusion

CROSS turns cross-rollout corrective evidence into recoverability-aware, localized supervision for language model reasoning. Corroboration selects repairs, guided recovery determines their eligibility and weight, and a frozen initial self-teacher supervises the original trajectory from the selected error boundary. CROSS achieves the highest cross-field average among the compared methods under independent best-checkpoint selection for each seed. Learning curves separately characterize token efficiency: selected mean-curve checkpoints combine higher scores with fewer supervised tokens than OPSD across the evaluated fields. Biology illustrates rapid learning from limited direct supervision. The results support jointly selecting corrective information and its supervision location as a promising route to more effective use of direct token supervision.

Limitations

Our evaluation covers Qwen3-4B on the science reasoning fields of SciKnowEval. The results characterize this model and task setting; broader language and reasoning capabilities require additional evaluation. We plan to extend the study to challenging mathematical reasoning datasets and base models at other parameter scales. These are future evaluation directions. The complete-system results also do not isolate the individual effects of corroboration, recovery weighting, and localization. Controlled ablations would clarify their respective contributions and interactions.

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A Online Training Algorithm

The full training loop implements the original-target objective in Section 3.

B Notation, Evidence Semantics, and Information Access

B.1 Notation and identity

Table 3 fixes symbols shared across the method. Algorithmic objects preserve the identities of their problem, target, upstream correction, and relevant peer or recovery attempts. Two corrections with identical text can remain distinct when their generating evidence differs. References carry provenance; they do not confer additional information access.

B.2 Information boundaries

Table 4 distinguishes explicit inputs to each model role. The task checker may use the reference answer without exposing it to other roles. Conversely, retaining a complete peer trajectory can expose an answer already present in that sampled trajectory; excluding a dataset-answer field does not imply answer-free textual evidence. The same distinction applies to correction text. Deterministic finding and localization introduce no additional semantic model call. Diagnostic and post-hoc labels do not participate in production decisions.

B.3 Availability and scientific outcomes

Runtime availability, completion, and task outcome are distinct axes. Available and complete rollouts with assessed incorrect or unparseable answers can become targets. Truncated raw rollouts and unavailable assessments do not enter the target or peer pools. Recovery has a different observation boundary: an available capped continuation is still assessed, and an unparseable answer is a behavioral failure in its denominator. Thus raw-rollout eligibility and recovery-attempt accounting must not share an undifferentiated failure flag.

An unavailable pair comparison casts no semantic vote. A missing exact token mapping prevents that correction from advancing. Identity or provenance inconsistency is a contract violation, rather than a negative task outcome or a semantic opposition vote. These distinctions preserve the populations to which each estimator and decision rule applies.

C Peer Construction and Correction Proposal Details

C.1 Lexical clustering and medoids

Peer normalization applies Unicode NFKC followed by case folding. Lexical tokens are word or number runs and individual non-whitespace symbols; token multiplicity is discarded in the set $\phi(z)$. All pairwise similarities are computed once. Thresholded similarities define an undirected graph, and its connected components provide the clusters. Connectivity is transitive: it does not assert that every pair within a component exceeds the threshold.

For a component $\mathcal{D}$ with more than one peer, its medoid is

$$z^{\star}_{\mathcal{D}} = \arg\max_{z_i \in \mathcal{D}} \frac{1}{|\mathcal{D}| - 1} \sum_{\substack{z_j \in \mathcal{D} \\ j \neq i}} s(z_i, z_j). \qquad (12)$$

A singleton selects its sole member. Ties and representative order follow the declared deterministic ordering protocol. Medoid selection reuses the similarity matrix, so n_p eligible peers require $n_p(n_p - 1)/2$ distinct similarity evaluations in total. Every resulting representative is retained. The value $\tau_{\text{lex}} = 0.8$ is a fixed setting; the available calibration does not establish its optimality.

C.2 Structure-aware reasoning units

The unitizer preserves exact substrings of the target. Protected code and mathematical spans remain atomic; paragraph, list, heading, and displayed formula structure supply boundaries. Prose is refined using sentence boundaries, standalone formulas remain units, and an ordinary single newline does not create a reasoning boundary. The same segmentation definition is used when resolving an anchor for localization. This keeps the proposed unit identity tied to its source character span.

C.3 Proposal sampling and bounded failures

The documented generator profile draws one proposal per target and peer pair with temperature 1.3, top-$p = 0.95$, top-$k = 20$, and additional thinking mode disabled. These are proposal-decoding settings, not vocabulary-support parameters for the distillation loss. If $L_{\text{tok}}(\mathbf{y})$ is the token count of the exact raw target text under the generator tokenizer, the output budget is

$$B(\mathbf{y}) = \max\left(64, \lceil 0.5 L_{\text{tok}}(\mathbf{y}) \rceil\right). \qquad (13)$$

Algorithm 1 Online original-target CROSS: one batch in five stages

Require: Initial student θ_0, ordered problem batches, G, M, τ_R, model roles and information profile
Initialize: freeze the teacher at θ_0 and initialize AdamW.
Repeat for each logical batch t: hold θ_t fixed throughout stages 1–4.

1: **Sample evidence.** For every problem x, sample G student rollouts and obtain typed task outcomes. Build the eligible failed-target pool and ordered peer representatives. Skip correction construction when fewer than two peer clusters remain.
2: **Select corrections.** For every eligible target $\mathbf{y}$, propose one correction per peer and group proposals by anchor. Apply Algorithm 2 to each group with at least two proposals; collect all selected corrections with their target provenance.
3: **Probe recovery.** For each selected correction c, resolve its exact prefix length k. Discard unmappable cases. Sample M guided continuations from θ_t and compute assessed attempts A and successes S. Retain the correction only if $A > 0$ and $S/A \geq \tau_R$.
4: **Assemble supervision.** Form $\mathcal{S}_t$ from all eligible original-target signals $(x, \mathbf{y}, c, k, S/A)$. Fix the complete signal collection and all loss denominators before accumulating any gradient.
5: **Commit the batch.** If $\mathcal{S}_t$ is nonempty, accumulate $\nabla_\theta \mathcal{L}_t$ at θ_t, apply one full-parameter AdamW update, and record signal consumption. Otherwise preserve student parameters and optimizer moments. Advance batch progress; the resulting student supplies the next batch's rollouts and recovery probes.

Table 3: Shared notation. Token sequences and suffix positions use zero-based indices.

Symbol	Meaning
t, θ_t	Logical training iteration and its student snapshot.
θ_0	Initial policy parameters, fixed for teacher evaluation.
x, G	Public task context and number of raw student rollouts per problem.
$\mathbf{y}, T_y$	Exact stored completion token sequence and its length.
$\mathcal{Z}, K$	Ordered peer representatives and the number of lexical clusters.
$u_1, \ldots, u_L$	Exact source reasoning units and their count.
$a, c, \mathcal{C}_a$	Proposed anchor, correction text, and indexed same-anchor proposal group.
$\mathcal{G}_\phi$	Proposal-model role; parameter sharing with other roles is not implied.
d_j^+, d_j^-, g_j	Direct support degree, opposition degree, and their difference.
$s^\star, k$	Proposed unit's character onset and retained original-token count.
P, U, Z	Exact retained prefix, original scored suffix, and suffix token count.
M, A, S, R	Planned recovery attempts, assessed attempts, successes, and S/A.
τ_R, R^2	Recovery eligibility threshold and signal-level weight.
$\mathcal{S}_t, N_t$	Indexed CrossSignal collection and its cardinality.
C_i^T, C_i^S	Teacher and student contexts for distillation.
$q_{i,u}, p_{i,u}$	Aligned teacher and student next-token distributions.

The complete input plus budget must fit the context window. The procedure does not crop target or peer evidence to force a fit.

Each pair receives a deterministic seed derived from a run-level generator seed, correction-case identity, and peer identity. This assigns randomness to logical pairs rather than their physical scheduling order. Structural parsing requires a valid proposed unit and non-empty correction text. Failed slots do not invalidate successful siblings. Exact duplicate corrections from different peers remain separate entries, and every non-empty anchor group is retained. Only the next stage imposes the two-proposal ingress requirement.

D Signed Corroboration Reducer

D.1 Semantic labels and canonical order

The relation model predicts the main actionable relation between two corrections at their shared proposed anchor. Wording or explanatory differences can still receive SUPPORT. OPPOSE requires incompatible main repairs. Different compatible repairs receive NOEVIDENCE when neither supports nor directly opposes the other. Relations are decoded greedily, one canonical orientation per unordered pair. In Eq. (3), $r_{j\ell}$ always refers to that stored canonical comparison; it is not a second reversed model call.

Table 4: Explicit role-specific information access. Causal trajectory history is separate from task-level privileged inputs.

Role	Model-visible scientific inputs	Excluded explicit inputs
Raw student	Public problem and its own causal history	Correction guidance and dataset answer fields
Proposal model	Public problem, complete numbered target, one complete peer	Separate dataset answer, checker diagnostics, post-hoc labels
Relation model	Public problem, target through the proposed unit, two corrections	Full peers, dataset answer, other relation judgments, recovery outcomes
Recovery student	Public task, selected correction, exact original prefix	Separate dataset answer, peer verification evidence, recovery statistics
Distillation student	Ordinary problem context and original causal token history	Explicit correction and dataset privileged information
Distillation teacher	Problem, correction, authorized dataset information, same causal history	Future tokens of the scored suffix
Objective checker	Task-specific reference information and the response to assess	No checker evidence is returned as additional model guidance

Algorithm 2 Same-anchor cross-path corroboration

Require: Ordered proposals $\mathcal{C}_a = (c_1, \ldots, c_n)$, $n \geq 2$, public task, target prefix through anchor
Initialize vertices $1, \ldots, n$, $E^+ = E^- = \emptyset$, assessed count $b = 0$, selected list $W = []$
for each unordered pair (j, ℓ), $j < \ell$, in canonical order **do**
  Evaluate its semantic relation once
  if the comparison is available **then**
    $b \leftarrow b + 1$
    if relation is SUPPORT **then**
      Add $\{j, \ell\}$ to E^+
    else if relation is OPPOSE **then**
      Add $\{j, \ell\}$ to E^-
    end if
  end if
end for
for each component $\mathcal{H}$ of $(\{1, \ldots, n\}, E^+ \cup E^-)$ **do**
  Compute d_j^+, d_j^-, g_j within $\mathcal{H}$
  $E_{\mathcal{H}} \leftarrow \{j \in \mathcal{H} : d_j^+ \geq 1\}$
  if $E_{\mathcal{H}} \neq \emptyset$ **then**
    Select $j^\star$ by Eq. (4)
    Append existing $c_{j^\star}$ and its provenance to W
  end if
end for
if $W \neq []$ **then**
  return W, ACCEPT
else if $b > 0$ **then**
  return [], REJECT
else
  return [], UNASSESSED
end if

D.2 Interpretation of the reducer

Both support and opposition connect vertices into relation cores. A core is therefore a region of observed semantic interaction, not necessarily a set of mutually agreeing corrections. Selection requires at least one direct support edge and maximizes signed support, then support degree, then stable order. It does not require the winning net score to be positive. Textual simplicity, length, minimality, and objective truth are not ranking variables.

A group may produce several representatives

from disconnected supported cores. If at least one representative exists, the group is accepted. If some comparisons are assessed but none yields a supported representative, the group is rejected. Only complete comparison unavailability produces UNASSESSED. Partial missingness is handled by the remaining observed relations. This is the complete acceptance logic used in the integrated method.

E Exact Token Localization

The target's original completion IDs are authoritative. Let $\widetilde{h}$ be their raw decoded text and h the canonical target text. The proposed unit supplies a character onset $s^\star$ in h. When canonicalization removes outer whitespace, the raw-coordinate onset is

$$\delta = |\widetilde{h}| - |\operatorname{lstrip}(\widetilde{h})|, \qquad \widetilde{s}^\star = s^\star + \delta. \quad (14)$$

The interval map is defined for the exact original token sequence. Invisible special tokens have no visible interval, and intervals can overlap when multiple byte-fallback IDs jointly contribute one Unicode character. Equation (5) selects the earliest contributor to the onset character.

At an exact token boundary, the prefix retains all preceding tokens. Inside a multi-character token, excluding that token may remove a small amount of text before the proposed onset. With overlapping intervals, choosing the earliest contributor avoids retaining part of the onset character. The amount of leftward retreat is diagnostic only; no retreat threshold changes acceptance. The procedure performs no retokenization of the desired text prefix, fuzzy matching, rightward projection, or new semantic localization.

A successful result adds only the prefix count k to the existing correction and target provenance. Recovery resolves $\mathbf{y}[:\ k]$ from the original IDs. Unavailable IDs or an unavailable exact mapping prevent advancement. Exact boundary safety is conditional on the inherited proposed unit; it does not establish that this unit is the true first error.

F Behavioral Recovery Protocol

F.1 Snapshot binding and continuation

The recovery model is bound to the snapshot that generated its target, not to a newer checkpoint available at execution time. It receives the non-privileged task context, selected correction, and authoritative assistant-prefix IDs. Only tokens newly generated after the complete input are recovery suffixes. The documented profile inherits the target rollout's sampling policy, changing only the per-attempt sample count, seed, and continuation budget. Its suffix budget is 4096 new tokens.

For the evaluated task interface, the required final answer can be checked from the suffix, so verification does not reconstruct the complete prefix-plus-suffix response. This scope must be reconsidered for tasks whose correctness depends on the entire response. Reaching the token cap is an observed continuation behavior: its available suffix is still evaluated.

F.2 Assessed denominator and gating

All assessed correct, incorrect, and unparseable attempts contribute to A; only correct attempts contribute to S. Explicit infrastructure unavailability contributes to neither. The score $R = S/A$ is consequently defined on assessed attempts. There is no minimum-assessed-count gate beyond $A > 0$ in the ratio-based method definition. The acceptance predicate $R \geq \tau_R$ and weight R^2 are separate operations.

The main experiments use $M = 7$ and $\tau_R = 4/7$. Eligibility is $A > 0$ and $S/A \geq 4/7$, including when fewer than seven attempts are assessed. For example, $S = 3, A = 5$ satisfies this rate threshold. The gate does not impose an additional fixed success-count requirement. Earlier recovery calibration is reported separately in Appendix I.

F.3 Per-attempt randomness

Attempt seeds are derived from a run-level recovery seed, localized-correction identity, and attempt index. The planned count M is not part of the slot identity. Increasing M therefore preserves the random-state assignments for existing logical slots. This supports nested-budget calibration without asserting bitwise equality across inference backends or hardware.

Recovery estimates task success under correction guidance. It does not include a matched unguided continuation control, so the measured rate alone does not identify the correction's causal improvement in success probability. This is an interpretation of the defined probe, not an additional filter or estimator.

G Objective Execution and Online Training

G.1 Causal alignment and full-vocabulary loss

For original-target scoring, suffix token $U_{i,u}$ is predicted from $(C_i^X, P_i, U_i[:u])$, where X denotes teacher or student. The current and future suffix tokens are not part of their own prediction contexts. Teacher and student task prompts can have different lengths, so alignment is by the common completion-token index rather than absolute prompt position. Let B_i^X be the number of serialized prompt tokens before the completion. With standard next-token logits, the prediction for $U_{i,u}$ is read at $B_i^X + k_i + u - 1$. This index relation follows directly from the zero-based sequence convention and causal next-token scoring.

For the shared vocabulary $\mathcal{V}$, the per-position divergence is

$$D_{i,u} = \sum_{v \in \mathcal{V}} q_{i,u}(v) \log \frac{q_{i,u}(v)}{p_{i,u}(v;\theta)}. \tag{15}$$

The stored continuation is teacher-forced, not regenerated. Full-suffix weighting applies to scored completion positions, while batch padding and prompt positions contribute no direct loss. The precise stored terminal-token and scoring-mask convention must be fixed by the experiment protocol. Distillation temperature is one. Teacher outputs, collected tokens, selection decisions, and measured recovery weights are fixed during the update.

G.2 Reduction and gradient partitioning

The coefficient of each supervised position in signal i is

$$\alpha_i = \frac{R_i^2}{N_t Z_i}. \tag{16}$$

Thus dividing the workload into physical microbatches must preserve the global N_t and each Z_i. If $\mathcal{B}_1, \ldots, \mathcal{B}_B$ partition the fixed signal index set, the same objective is

$$\mathcal{L}_t(\theta) = \sum_{b=1}^{B} \sum_{i \in \mathcal{B}_b} \alpha_i \sum_{u=0}^{Z_i - 1} D_{i,u}. \tag{17}$$

Equal averaging of independently normalized microbatch losses would generally change these coefficients when their signal counts differ.

As a direct algebraic consequence of forward KL with fixed teacher targets and temperature-one softmax logits $\ell_{i,u,v}$,

$$\frac{\partial \mathcal{L}_t}{\partial \ell_{i,u,v}} = \alpha_i \big(p_{i,u}(v) - q_{i,u}(v)\big). \tag{18}$$

This is the gradient identity used in Eq. (10). Prefix positions have no direct distillation term, while suffix losses can backpropagate through prefix representations.

Equation (11) in the main text gives the resulting signal and batch coefficient sums. It separates the recovery-dependent loss weights from the number of directly supervised token uses.

G.3 Logical updates and state continuation

All valid signals from a completed problem batch are normalized together. The recorded Phase B implementation used batches of 16 problems; the final partial batch was processed with its actual inputs. For any positive signal count, the same signal-level reduction applies. A legitimate zero-signal batch advances data progress without changing model parameters or optimizer moments. An incomplete transaction causes no partial optimizer update; a fully processed final data batch may contain fewer than 16 problems.

The original-target online loop is shown in Algorithm 1. The teacher stays at θ_0, while raw sampling and recovery use the batch's frozen student snapshot. An already committed signal is not replayed in a later committed update. Training-state continuation preserves the student, optimizer, random states, data progress, and signal-consumption identity. The earlier qualified adapter update and the later full-parameter update are different parameter configurations of the same objective; the main text describes the latter.

H Documented Trajectory and Privilege Variants

H.1 Recovery-repaired trajectory collection

Original-target scoring uses the singleton trajectory collection $\mathcal{U}_i^{\text{orig}} = (\mathbf{y}_i[k_i:])$. A separately defined variant retains all assessed-correct suffixes from the same recovery execution that produced R_i. For successful attempt indices $\mathcal{J}_i^+$, define the indexed family

$$\mathcal{U}_i^{\text{rec}} = \big(\widetilde{U}_i^{(m)}\big)_{m \in \mathcal{J}_i^+}. \tag{19}$$

Every suffix shares the original exact prefix P_i. The indexing preserves multiplicity, including identical texts from different successful attempts. Capped-but-correct suffixes are retained. Collection is deterministic: no new recovery generation, representative sampling, or fallback is performed. A different recovery execution cannot replace the bound evidence merely because its aggregate rate is the same.

All these continuations belong to one correction-level CrossSignal. Their number does not multiply the count of independently weighted signals, and R_i^2 remains a single signal-level factor. This is an exploratory trajectory construction and is not part of the reported main method or its empirical claims; the main loss and training loop use original-target scoring. A complete repaired-trajectory objective additionally requires a declared within-signal reduction over its continuations.

H.2 Dataset privileged information

The documented teacher profiles are: no dataset privilege, answer-only privilege, and reference-trace privilege containing both answer and reference solution. The selected correction is a separate source of teacher guidance in all three profiles. Student distillation inputs remain ordinary problem context plus causal token history. All reported experiments in Chemistry, Biology, Physics, and Materials Science use answer-only privilege.

I Development Diagnostics and Reporting Scope

The following observations motivated or checked individual design choices. They come from separate development populations and protocols and must not be combined into a single end-to-end funnel or treated as final training results.

I.1 Peer evidence and proposal coverage

A rollout diagnostic at $G = 16$ and sampling temperature 1.3 reported about 8.03 outcome-correct trajectories and 7.84 distinct correct clusters per problem. Separate reference-quality reviews reported reviewer-valid precision of approximately 27.6% without the earlier reference judge and 34.9% with the compared reference-judge configuration. These observations motivate retaining multiple peers without treating their reasoning as certified truth.

A joint proposal procedure produced usable outputs for 69 of 76 pairs, versus 48 of 76 and 49 of 76 for two factorized locator-then-patcher configurations. A later 48-question diagnostic produced 1395 structurally valid candidates from 1444 pair slots, and 91.0% of target cases had at least one anchor group with two or more proposals. Structural coverage does not measure correction correctness. The corresponding review still identified locality and excessive breadth as concerns.

I.2 Relation judgments and ordering sensitivity

A verifier run assessed all 2018 planned pairs across 312 cases and selected 260 corrections. A separate semantic audit reported reviewer agreement of about 69.4% on production SUPPORT and 50.0% on production OPPOSE. Swapping pair order yielded 87.2% pair-label agreement and 92.6% case-decision agreement. A bidirectional two-vote counterfactual increased the selected count from 260 to 274 at twice the comparison-call count. These diagnostics characterize a noisy, order-sensitive corroboration procedure; they do not establish oracle verification or justify silently replacing the canonical single-comparison rule.

I.3 Localization and recovery calibration

Localization succeeded for 260 of 260 corrections: 244 matched exact token boundaries and 16 retreated by one character. Maximum retreat was one character, with zero right crossings. These measurements validate the projection against proposed anchors, without auditing whether those anchors were semantically correct.

Historical recovery calibration, separate from the main $M = 7$, $\tau_R = 4/7$ configuration, used 130 corrections from 29 problems and recorded 2080 assessed attempts. Eligibility with $M = 12$ agreed with $M = 16$ on 96.9% of corrections. Relative to the reported original student accuracy of 54.5%, $7/12$ was the lowest attainable twelve-assessed-attempt fraction above that value. Squared recovery weighting was reported to retain an effective-sample-size ratio of about 90.4%. These are profile-specific calibration observations, not evidence that the same budget or threshold is optimal across tasks. The original accuracy also does not replace a matched unguided recovery control.

I.4 Implementation validation versus model improvement

An earlier adapter-training qualification processed 50 problems in four updates and consumed 102 signals, including a two-problem tail batch. A later runtime qualification completed 50 consecutive updates. These are separate runs with different parameter configurations. A repaired-trajectory canary retained all nine successful suffixes and replayed their evidence without new generation. Such results support objective execution, evidence preservation, and state continuation. They do not establish out-of-domain generalization, superiority to a baseline, or the necessity of each module. Observations of output contraction and sparse or empty signal batches require separate training and evaluation analyses.

J Extended Related Work

J.1 Rationale Learning and Privileged Information

Chain-of-Thought prompting introduces intermediate reasoning demonstrations into the inference context without updating model parameters (Wei et al., 2022). STaR turns generated rationales into training data: it retains answers that pass task verification, uses answer-conditioned rationalization for initially unsuccessful examples, and fine-tunes on the resulting successful reasoning traces (Zelikman et al., 2022). Distilling Step-by-Step uses teacher-generated rationales as auxiliary targets in a multi-task objective combining label prediction and rationale generation (Hsieh et al., 2023). These works establish natural-language reasoning as both a prompting resource and a source of supervision. Their central training objects are demonstrations or generated reasoning sequences; CROSS instead selects corrections associated with errors in student-generated trajectories.

Generalized distillation connects knowledge transfer with learning from privileged information, allowing a teacher to exploit representations unavailable to the student at test time (Lopez-Paz et al., 2015). This provides a conceptual foundation for asymmetric teacher and student contexts. MiniLLM studies sequence-level reverse KL and develops policy-based optimization for generative distillation (Gu et al., 2024). GKD studies student-generated training outputs and alternative divergences, separating the source of training trajectories from the choice of distribution-matching objective (Agarwal et al., 2024). These dimensions remain relevant when the teacher is instantiated through privileged self-conditioning.

J.2 Self-Teaching Contexts and Multi-Rollout Evidence

OPSD evaluates student rollouts under both question-only and reference-conditioned policies, obtaining soft supervision from their predictive distributions (Zhao et al., 2026a). SDPO extends self-teaching to rich environment feedback and also uses successful attempts as feedback when the environment returns only scalar outcomes (Hübotter et al., 2026). SDFT conditions the teacher on demonstrations and studies acquiring new skills or knowledge while retaining earlier capabilities (Shenfeld et al., 2026). These studies differ in the information available during training and in their evaluation objectives. They motivate distinguishing the origin of a teaching signal from the mechanism that determines its use.

For unsuccessful trajectories, ROSD pairs the target with a shortest successful rollout from the same group and produces a corrective idea together with an error quote (Zhao et al., 2026b). The quote determines the suffix receiving direct distillation supervision. ROSD also provides positive reflections for successful rollouts; its training is consequently broader than correction of failed trajectories alone. MOPD explores both successful-peer conditioning and contrastive contexts containing successful and failed peers, with the target excluded from its own peer evidence (Yu et al., 2026b). OGLS-SD aggregates logits induced by positive and negative guidance contexts and uses their difference to steer the target distribution (Yang et al., 2026). These methods place comparative evidence in reflection text, teacher context, or numerical targets.

GATES introduces a question-level consensus gate under asymmetric document access (Stein et al., 2026). Multiple document-conditioned tutor answers determine a consensus pseudo-label, and questions lacking sufficient agreement are skipped. The framework supports both tutor-trajectory distillation and on-policy updates under the same gate. CROSS operates on a different evidence object: proposals for repairing the same localized error. Cross-path corroboration concerns the compatibility of these proposed repairs. Agreement on a repair and agreement on a final answer therefore provide distinct forms of evidence, and nei-

ther constitutes an independent guarantee of reasoning correctness.

J.3 Feedback Utility, Process Supervision, and Continuation Outcomes

SCoRe trains multi-turn self-correction through on-policy reinforcement learning, addressing distribution shift in offline correction data and behavior collapse toward uninformative revisions (Kumar et al., 2025). CTRL optimizes a critic according to the quality of revisions produced by a fixed generator after receiving its feedback (Xie et al., 2025). This makes feedback quality operational through downstream behavior. The respective learning targets are a self-correcting policy and a useful critic. CROSS uses behavioral evidence to select the corrective information supplied during distillation.

Process supervision addresses a related question at the reasoning-step level. Lightman et al. (2024) compare outcome-supervised and process-supervised reward models, with human judgments of intermediate steps providing the process labels. Math-Shepherd automates process annotation by sampling completions from a partial solution (Wang et al., 2024). Its hard and soft annotations reflect, respectively, whether any sampled completion succeeds and the fraction of successful completions. VinePPO uses the ability to restart generation from an intermediate prefix to estimate expected returns through Monte Carlo continuation, replacing learned value estimates in credit assignment (Kazemnejad et al., 2024).

The role of a continuation outcome depends on its conditioning and downstream use. In CROSS, recovery is measured for a particular prefix, a selected correction, and the student snapshot responsible for the original rollout. It is a conditional task-success statistic used for supervision admission and weighting. It does not assign correctness labels to every reasoning step or directly estimate a policy-gradient advantage. A high recovery rate also does not, by itself, isolate the improvement attributable to the correction relative to continuation without that correction. This distinction separates the behavioral selection criterion from both process annotation and causal claims about correction utility.

J.4 Distributional and Token-Level Supervision Selection

Purified OPSD contrasts a question-and-reference teacher with a reference-only teacher and combines the resulting likelihood ratio with a base distribution to construct a purified target (Shen et al., 2026). Its central intervention concerns the distribution being distilled. PW-OPSD diagnoses teacher-token reliability by forcing candidate alternatives on sampled reasoning prefixes and measuring subsequent student-context completion success (Liu et al., 2026). The reported diagnostic focuses on a selected correct-trajectory subset; the training method then uses increasing position-dependent weights. This distinguishes a behavioral diagnostic that motivates a weighting rule from an online test of each proposed correction.

EDGE-OPD combines privileged-context-guided sampling with a mask based on how much the privileged context increases a sampled token's probability (Lazaridis et al., 2026). Its identity and mathematical-reasoning experiments show that the usefulness of this evidence depends on what information the context supplies. A probability increase under privileged context can support internalizing background information while still emphasizing answer-specific behavior in reasoning tasks. This illustrates why a distributional evidence score and task-relevant learning value should be distinguished.

TIP organizes token importance through student entropy and teacher-student divergence, including confident predictions that nevertheless disagree strongly with the teacher (Xu et al., 2026). Its selection rules retain informative positions for distillation instead of supervising every rollout token. SEAD uses joint teacher-student entropy to select positions and assign divergences, while also scheduling the divergence mixture and restricting the prompt distribution through a competence-based curriculum (Lee et al., 2026). CROSS selects target-bound corrections using cross-path relations and conditional recovery, then localizes their supervision to a suffix. Thus, its selectivity acts through the semantic unit being taught and the region where it is applied; retained suffix positions can still receive dense distributional targets.

J.5 Reward-Driven Optimization and Efficiency Comparisons

DAPO studies scalable reward-driven optimization through asymmetric clipping, dynamic sampling, token-level loss aggregation, and overlong reward shaping (Yu et al., 2026a). Its token-level policy-gradient loss changes the relative influence of responses of different lengths, while its sampling rule removes groups without informative reward variation. These interventions concern optimization and sample utilization, providing context for interpreting strong reinforcement-learning baselines without making correction selection their objective.

SERL connects hindsight information with reward-driven updates in multi-turn agents (Li et al., 2026). A hindsight-conditioned teacher produces a bounded, sign-aware reweighting of the reward-derived advantage, and teacher-based modulation is restricted to executable action spans. Reward determines the update direction, while hindsight affects its magnitude and placement. This differs from directly matching a selected privileged teacher distribution and illustrates that teacher information can affect several distinct parts of a learning objective.

Efficiency claims also depend on the resource being counted. OPSD evaluates performance against generated-token usage (Zhao et al., 2026a), while SDPO examines generation efficiency and training time (Hübotter et al., 2026). TIP and SEAD directly investigate which token positions receive distillation supervision (Xu et al., 2026; Lee et al., 2026). These are related but distinct measurements. Fewer directly supervised positions can coexist with additional sampling or scoring work to obtain those positions. Moreover, a distributional distillation target and a reward-derived policy-gradient term supply different kinds of information at a token position. For CROSS, the distinction between supervision-token efficiency and end-to-end efficiency is particularly relevant because correction construction, verification, and recovery incur work before the selected suffix is distilled. Supervised positions, generated tokens, and training cost therefore characterize complementary aspects of learning efficiency.

J.6 Comparison by Evidence and Selection Unit

Table 5 summarizes representative mechanisms according to their selection or aggregation unit, evidence source, and role in learning. It compares methodological interfaces rather than ranking their empirical performance.

K Additional Experimental Protocols and Training Curves

Table 1 reports per-seed best-checkpoint performance. Table 2 reports supervision exposure at selected mean-curve checkpoints. The figures show the full three-seed mean trajectories. This appendix details the reporting protocols and additional curves.

K.1 Checkpoint selection and training populations

All methods use seeds 42, 43, and 44. After training, the checkpoint with the highest test in-domain mean@16 is selected independently for each seed. Table 1 reports the mean and sample standard deviation of the resulting three best scores. Its four-field average equally weights the domain means; no average is reported when a domain result is unavailable. CROSS processes 500 batches of 16 questions and evaluates every 20 batches. Baselines run for 200 optimizer steps with 16 questions per step; their recorded evaluation steps are $\{0, 40, 80, 120, 160, 200\}$.

CROSS and all baselines use the same fixed 90/10 train/test split with split seed 42, system prompt, and V2 answer parser. The Materials training set contains 840 questions after one exclusion; the item identifier is given in Appendix K.4.

K.2 Supervised-token accounting

A supervised token use is one valid token position included in the objective of one committed optimizer update. If a target position occurs in several correction-level signals, each occurrence is counted. Reuse in another update also counts again. Prompt and masked prefix positions, padding, empty signals, and batches without an optimizer update contribute zero. Recovery weights scale losses and do not fractionally discount token counts.

For the original-target, uniform-full-suffix objective, the count for a committed update is

Table 5: Evidence and selection units in related distillation methods. The entries describe the principal mechanism relevant to CROSS; they are not exhaustive summaries of each training system.

Method	Evidence and unit	Role in learning
ROSD (Zhao et al., 2026b)	Reflection and error quote for a target rollout.	Reflection-conditioned teacher and localized suffix supervision.
MOPD (Yu et al., 2026b)	Successful and failed peers from a rollout group.	Comparative teacher-context construction.
OGLS-SD (Yang et al., 2026)	Aggregated outcome-conditioned logits at a target prefix.	Steering of the distillation target.
GATES (Stein et al., 2026)	Agreement among privileged-context tutor answers.	Question-level gating, with trajectory eligibility for off-policy losses.
PW-OPSD (Liu et al., 2026)	Token position informed by forced-continuation diagnostics.	Position-dependent distillation weighting.
EDGE-OPD (Lazaridis et al., 2026)	Context-induced probability change at a sampled token.	Evidence masking with guided sampling.
TIP (Xu et al., 2026)	Student entropy and teacher-student disagreement at a token.	Selection of directly supervised positions.
SEAD (Lee et al., 2026)	Joint token entropy and prompt-level competence.	Token selection, divergence scheduling, and curriculum.
CROSS	Same-error correction corroboration and conditional student recovery.	Correction admission and weighting with localized distillation.

$\sum_{i\in\mathcal{S}_t} Z_i$, with Z_i restricted to valid directly supervised positions. The curves report cumulative supervision recorded for each method's training objective. These token counts characterize direct training exposure; they do not include the generation or scoring needed to construct the supervision signals.

$$T_{\mathrm{sup}}(c) = \sum_{t\le c:\ \text{update committed}} \sum_{i\in\mathcal{S}_t} Z_i. \tag{20}$$

K.3 Performance versus cumulative supervision

Figure 2 and Figures 3–5 show in-domain mean@16 against cumulative supervised tokens. Every method uses the same three seeds, 42, 43, and 44. The curves summarize mean performance at corresponding cumulative token budgets. Shaded bands represent one standard deviation across seeds.

This trajectory summary is distinct from averaging independently selected best checkpoint scores. Writing $y_s(c)$ for the observed ID score of seed s at checkpoint c, the table reports

$$\frac{1}{3}\sum_{s\in\{42,43,44\}} \max_c y_s(c). \tag{21}$$

The curves instead average seed-level performance as a function of cumulative token exposure. Different seeds can reach their best scores at different token budgets, so the table mean need not appear on the mean trajectory. The figures characterize token efficiency independently of those selected best-score summaries.

Table 2 uses a separate selection over the checkpoint rows of each three-seed mean curve, restricted to positive training steps. It reports cumulative tokens at the highest curve score, with ties resolved toward fewer tokens. A dash is reported if no such checkpoint recovers the initial score. The OPSD/CROSS ratios compare the token counts at these selected curve checkpoints. Per-seed best-score selection in Table 1 and mean-curve selection in Table 2 are computed separately.

K.4 Documented runtime configuration

The supplied runtime export identifies Qwen/Qwen3-4B revision `1cfa9a7208912126459214e8b04321603b3df60c` for the student and frozen model roles, including the proposal and relation models. It records 16 raw rollouts per problem, an 8192-token cap for rollout, recovery, and evaluation, and a 32-token cap for greedy relation judgments. The generator's dynamic limit is given in Eq. (13). AdamW uses learning rate 10^{-5}, $\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$, and zero weight decay, with no scheduler, warmup, or gradient clipping. Student parameters are FP32 with BF16 autocast; teacher scoring is BF16 and gradient-free. The science adapter provides answer-only teacher privilege for all four fields. These are the documented shared runtime settings; per-run resolved man-

ifests remain necessary for complete execution provenance.

The dataset export identifies SciKnowEval revision `92ef969ad0a8bd6e195e0ac18af2c46e307e0cc2`, restricted to L3 two- or four-option questions and split with seed 42. Training cycles through a fixed content-hash order. The excluded Materials training item is `sciknoweval-material-00368`; the evaluation set is unchanged.

Asset versions and licenses. The model and tokenizer are `Qwen/Qwen3-4B` at the revision above, released under Apache-2.0. The dataset is SciKnowEval V2 (`hicai-zju/SciKnowEval`) at the revision above, released under MIT. These identifiers and upstream-declared licenses are recorded in the third-party asset inventory for this study.

K.5 Intended use and risks

These experiments study supervision for language model reasoning on benchmark questions. Outcome checks do not establish that generated explanations or corrections are factually sound at every step. Reusing incorrect corrective feedback could reinforce errors, so benchmark gains should not be treated as validation for scientific decision-making. Applications require evaluation in the intended domain and appropriate expert oversight. The reported token counts also omit generation and teacher computation and do not measure environmental impact.

L Documented Prompt Templates

The supplied runtime export provides the following literal message templates. Braced placeholders are populated by the saved problem and trajectory data. The role labels `SYSTEM:` and `USER:` identify messages and are not part of their content. The tokenizer applies the Qwen chat template with thinking disabled. The task prompt contains the question, a blank line, labelled options, and the instruction "Please reason step by step." These shared templates document the runtime profile; per-run rendered prompts and token bindings require the corresponding saved records.

L.1 Student rollout and evaluation

```
You are solving a multiple-choice science
    problem. Reason carefully, then give the
    final answer as exactly one option letter
    inside an <answer>...</answer> tag, for
    example <answer>B</answer>.
```

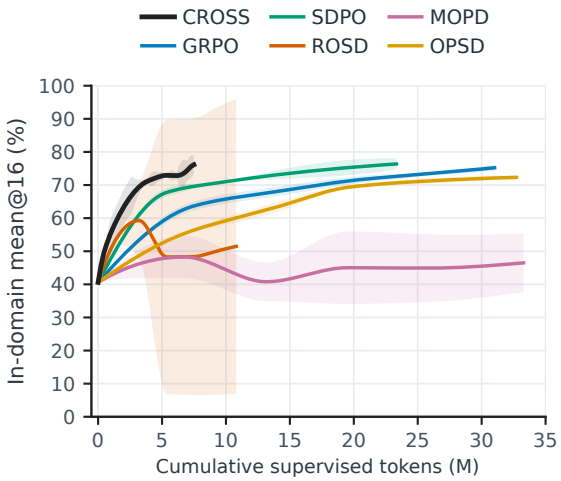


Figure 3: Chemistry in-domain mean@16 versus cumulative supervised tokens. Curves show means over seeds 42, 43, and 44 at corresponding token budgets, separately from the per-seed best scores in Table 1. Shaded bands represent one standard deviation across seeds.

L.2 Correction proposal

```
SYSTEM: You are a careful tutor. Respond
    strictly in the required format and nothing
    else.
USER:
[Problem]
{public_problem.prompt}

[Target rollout]
{numbered_target_units}

[Peer rollout]
{peer_rollout_text}

Find the first target unit containing an
    incorrect reasoning step.
Use the peer rollout as evidence to write a
    local correction for the same error.

Respond in this format:
{
  "first_error_unit": 1,
  "correction_text": "..."
}
```

L.3 Pairwise relation

```
SYSTEM: Compare two proposed corrections for the
     same point in the target reasoning.
Return exactly one JSON object and nothing else.
USER:
[Problem]
{public_problem.prompt}

[Target reasoning up to the point being
    corrected]
{prefix}

[Correction A]
```

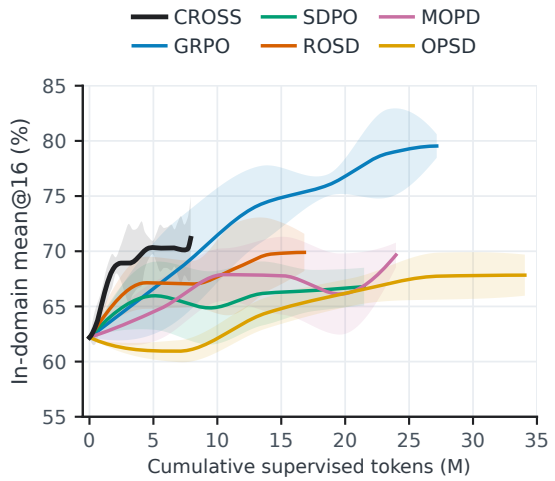


Figure 4: Physics in-domain mean@16 versus cumulative supervised tokens. Curves show means over seeds 42, 43, and 44 at corresponding token budgets, separately from the per-seed best scores in Table 1. Shaded bands represent one standard deviation across seeds.

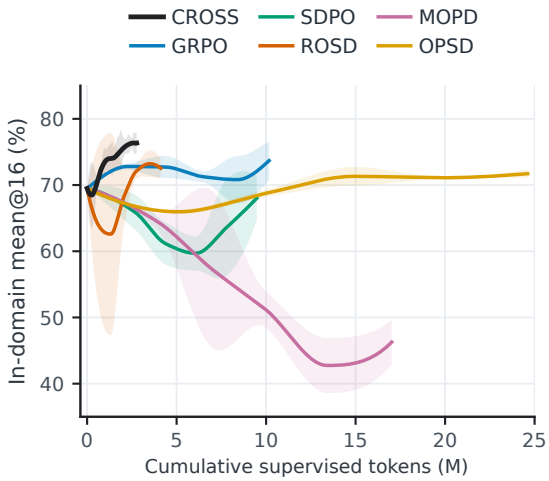


Figure 5: Materials Science in-domain mean@16 versus cumulative supervised tokens. Curves show means over seeds 42, 43, and 44 at corresponding token budgets, separately from the per-seed best scores in Table 1. Shaded bands represent one standard deviation across seeds.

```
{correction_a}

[Correction B]
{correction_b}

Classify the relation between the two
    corrections:

SUPPORT:
They propose the same main actionable repair at
    this point in the reasoning.
Their wording and amount of explanation may
    differ.

OPPOSE:
They propose incompatible main actionable
    repairs at this point in the reasoning.

NO_EVIDENCE:
They do not directly support or oppose each
    other,
or their relation cannot be determined reliably.

Judge only the relation between the corrections.

Respond exactly as one of:

{"relation":"SUPPORT"}
{"relation":"OPPOSE"}
{"relation":"NO_EVIDENCE"}
```

L.4 Guided recovery

```
[Verified correction]
{correction_text}

Use this correction as guidance.
Continue directly from the exact assistant
    prefix supplied after this message.
```

L.5 Answer-only teacher context

```
[Dataset answer]
{reference_answer}

[Verified correction]
{correction_text}

Use the supplied evidence to solve the original
    problem. Use the verified correction to
    guide continuation of the supplied assistant
     trajectory. Preserve valid reasoning and
    follow the original answer format.
```